

	Case Name: Beer Tasting	Sector	Event Management	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule Risk Analysis	Schedule with resources Schedule with costs Random simulation One of nine std. scenarios User defined distributions	
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Date	2015			
File Name	C2015-25 Beer Tasting		Project Control Automatic tracking Tracking based on user input	

1. Project description

Project authenticity

A Beer Tasting event involving detailed preparations, telechecks, material deliveries, tastings, and clean-up processes across multiple locations over several days.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	26	Serial/Parallel (SP)	16%
Planned Duration (PD)	14 days*	Activity Distribution (AD)	40%
Budget At Completion (BAC)	1.210 €	Length of Arcs (LA)	21%
Renewable Resources	4	Topological Float (TF)	19%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity				Time sensitivity		
	avg [%]	std dev [%]	skew [-]		avg [%]	std dev [%]	skew [-]
CRI-r	10.0	15.8	-1.1	CI	31.4	26.6	-0.3
CRI-rho	45.5	7.7	1.2	SI	40.8	34.8	-0.0
CRI-tau	73.0	41.6	0.9	SSI	11.3	20.4	-1.9
				CRI-r	6.4	13.0	-2.2
				CRI-rho	44.5	14.9	1.5
				CRI-tau	78.2	34.8	1.2
	Resource sensitivity						
	avg [%]	std dev [%]	skew [-]				
CRI-r	4.5	17.1	-3.7				
CRI-rho	4.5	17.1	-3.7				
CRI-tau	4.5	17.1	-3.7				

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	265.6	-265.6	1	0.9	0.7
PV - SPI	179.4	-179.4	CPI	1.0	0.8
PV - SCI	174.4	-174.4	SPI	7.5	7.5
ED - 1	869.5	-869.5	SPI(t)	8.7	8.7
ED - SPI	179.4	-179.4	SCI	7.5	7.5
ED - SCI	175.9	-175.9	SCI(t)	8.8	8.8
ES - 1	192.1	-192.1	0.8 CPI + 0.2 SPI	4.5	4.5
ES - SPI(t)	30.6	3.4	0.8 CPI + 0.2 SPI(t)	7.2	7.2
ES - SCI(t)	30.7	3.1			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 3 tracking periods with a length of approximately 1 week. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-470.0	0.0	-2.7	0.7	1.0	1.0	1.0
std dev	173.2	0.0	4.6	0.0	0.0	0.0	0.0
final	-570.0	0.0	-8.0	0.7	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	14 days
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Real Duration	15 days
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Early	7.1%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	13.0	1.0	2.8	-2.8
PV - SPI	14.0	1.0	5.6	-5.6
PV - SCI	14.7	0.6	7.4	-7.4
ED - 1	15.0	0.0	8.3	-8.3
ED - SPI	15.3	0.6	9.3	-9.3
ED - SCI	16.0	1.0	11.1	-11.1
ES - 1	17.0	1.0	13.9	-13.9
ES - SPI(t)	18.0	1.0	16.7	-16.7
ES - SCI(t)	19.0	1.0	19.4	-19.4

2.3.3.3. Cost forecasting

BAC	1.210 €
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Real Cost	1.780 €
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Over Budget	47.1%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	1680.0	173.2	1.9	1.9
CPI	1853.6	64.1	1.4	-1.4
SPI	1680.0	173.2	1.9	1.9
SPI(t)	1680.0	173.2	1.9	1.9
SCI	1853.6	64.1	1.4	-1.4
SCI(t)	1853.6	64.1	1.4	-1.4
0.8 CPI + 0.2 SPI	1804.7	42.7	0.5	-0.5
0.8 CPI + 0.2 SPI(t)	1804.7	42.7	0.5	-0.5